LS-CAT's Plan for APS-U Dark Period Data Collection

LS-CAT Users Meeting Monday 2/27/23

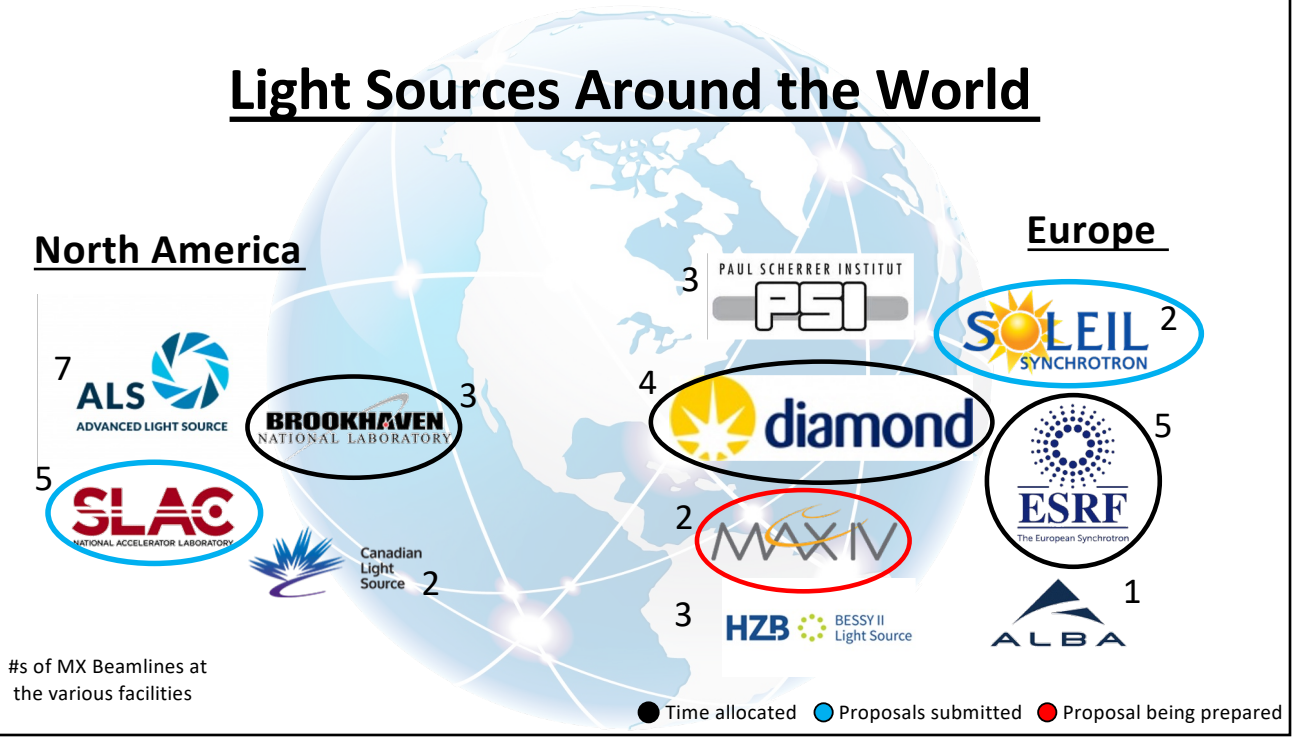
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Master Plan for Dark Period Data Collection

- Survey all Members about their current Research Projects and Grants
- Write and submit Block Allocation Group (BAG)/Project/Rapid Access proposals for the various lightsources around the world
- Awarded beamtime allocations would be based on CAT Member shares
- CAT Member Institutions will internally schedule the time in much the same way as they currently do for LS-CAT time
- The total amount of time we will receive is not yet known, as proposals are still being written (1), and reviewed (3)

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Light Sources Around the World



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BAG Proposals (Submitted last Fall)

- ESRF MX-2480 (2-year duration, 2 runs/year)
  - 40 shifts granted for 1st Run, 1 March - 30 July 2023
  - 2nd run 28 August - 10 December 2023
  - 1st visit 02-03 March 2023 – 2 shifts
- NSLS-II BAG-311288 (2-year duration, 3 runs/year)
  - 10 shifts granted for 1st Run, 1 March – 5 April 2023
  - 2nd Run May-August 2023, requested 20 shifts
  - 3rd Run Sept-December 2023
  - Not yet sure of our annual allocation
- Diamond Light Source - Arrangement with APS superseded our Proposal
  - Up to 12.5% of time on 4 beamlines granted to all APS MX Users
  - Includes SBC, LS, SER, GM/CA and NE-CATs (GUs and CAT Members)







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Arrangements by the APS





- Diamond Light Source -
  - Up to 12.5% of time on 4 beamlines granted to all APX MX Users
  - Includes SBC, LS, SER, GM/CA and NE-CATs (CAT Members, GUs)

- The 'Formula' being used to determine how the Diamond time will be shared between CAT Members and GUs
- It was rounded to a '40:60 split'
- Only 16 GUs applied
- LS-CAT Members' share of this beamtime in hours/month, is currently still TBD and will be decided via the APS MX-BAOC, each CAT has 1 Member (Beamtime Allocation and Oversight Committee)

Sector	"# BL"	CAT	GU
SBC	200	0	18
LS	300	20	7
SER	200	14	5
GM/CA	200	2	16
NE	200	2	16
Totals	1100	38	62

- Stanford Synchrotron Radiation Lightsource -
  - APS is trying to establish an arrangement with SSRL, but so far is only being offered beamline 14-1, a BM Line
  - BAG Proposals we submitted allowed choosing 4 beamlines
  - No timeline in place with the APS agreement
  - And we wait to hear about our BAG Proposals

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Necessary Hardware

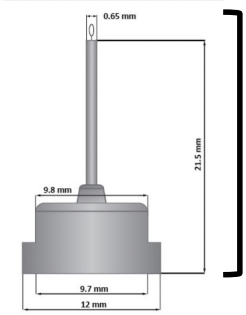
- SPINE bases w/ base-to-pin length of 22 mm (all except SSRL)
  - This is the format most already use at LS-CAT
  - Samples with incorrect bases will not be sent
  - Maximum loop size ~700 μm (NSLS-II)
- Unipucks
  - LS-CAT has purchased 5 new sets for use with your samples during the APS-U (7)
  - This will be our new format after the Dark Period, so start planning/budgeting
  - We will move your samples sent to us in your EMBL baskets to our Unipucks for shipping
- LS-CAT has purchased 2 new dry shippers/cases (4)

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Necessary Hardware

SPINE Bases –





22 mm from bottom of base to tip of loop





Unipucks – custom colors available, and get them engraved with labels humans can read, and also 2-D barcodes (all free)

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Logistics & Sample Handling for Data Collection

- Schedule beamtime at a constant interval
  - Publish a beamtime calendar
  - Implement an online portal to request time
- LS-CAT will consolidate User's samples into our Unipucks (Dry Shippers hold 7 pucks, 16 samples each)
- Ship samples to and from the lightsources through LS-CAT, then return bases/pins to users
  - International shipments via FedEx will be coordinated via NU's Customs Broker (Priority FedEx takes 2 days)
- File all Safety Forms, After Run Summary forms, and Reports







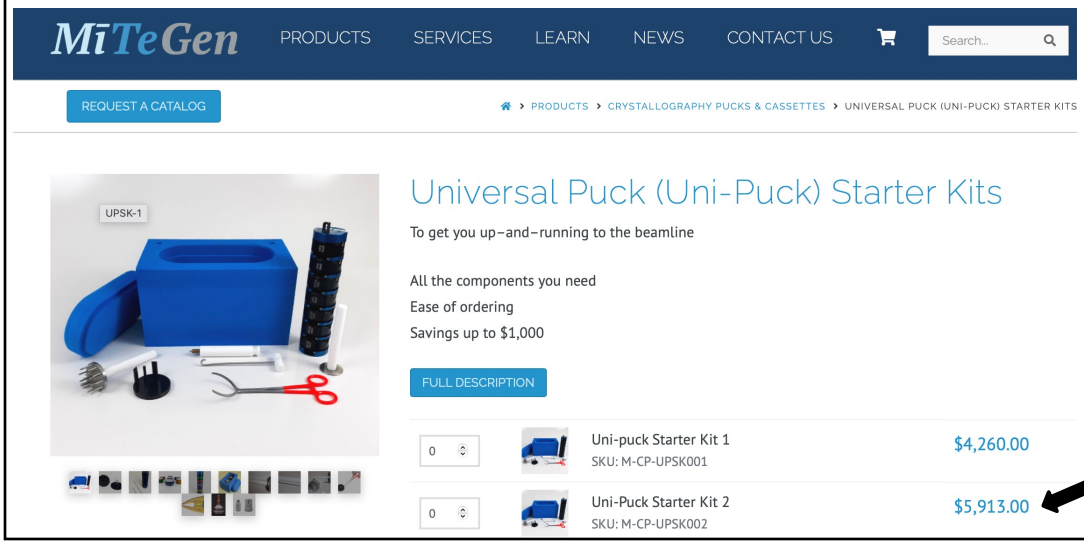
(SSRL casette)



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“Unipuck Starter Kit”, needed **after** the APS-U

There are several vendors, this one looks to be the least expensive. You don’t need to purchase these for a while. We have plenty for the Dark Period, and will transfer your samples into them for experiments



Note:
Puck holder may not fit into VWR Dry Shippers (?), but do fit into Taylor Wharton CX100s.

This second option includes a Dry Shipper and Case

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Data Workflow at LS-CAT

- Implement ISPyB at LS-CAT (Information System for Protein Crystallography Beamline)
  - Creation of sample shipments and tracking to the beamline
  - Management of sample, crystal and protein information
  - Real time monitoring of your data collections (diffraction images and crystal snapshots, harvesting of output from automated data reduction programs)
  - View historical experiment and data analysis information
  - Create and edit experiment reports
  - Export your experiment's meta-data back to your home lab
- Users will download data to home via Globus from LS-CAT, as usual
  - Coordinated via ESAF numbering scheme
- Implement post-run data processing, as desired
- Implement initial structure solution, as desired

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First Trial Run – Data Collection at NSLS-II 11/29/22





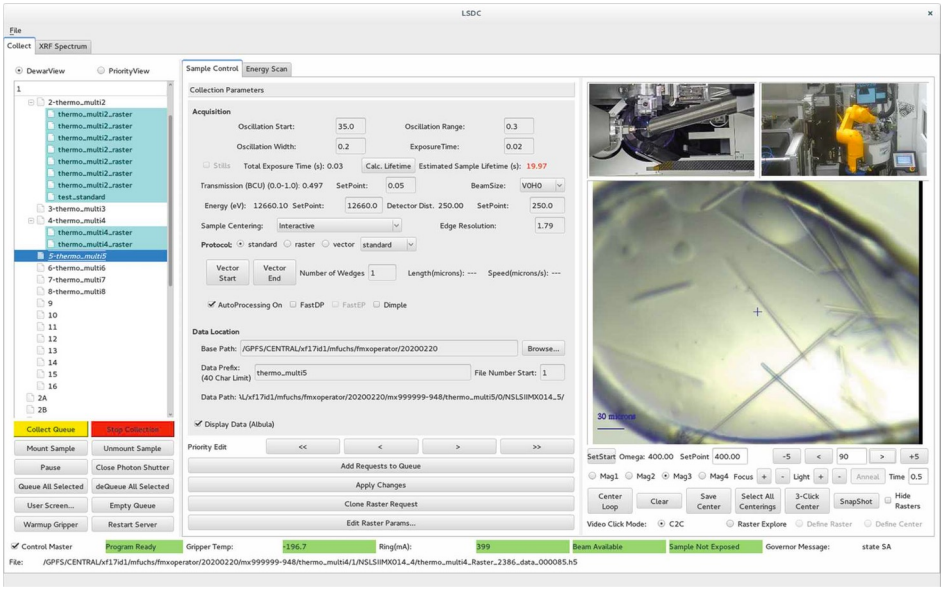




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Overview of User Operations at FMX 17-2

- Load sample spreadsheet
  - Puck name
  - Position
  - Sample name
  - Proposal ID
- Raster screening to find crystal
  - Adjustable grid sampling
  - Automated
  - Very fast
  - Accurate
- Vector data collection
- Quickly change focused beam size between 5x3µm & 10x10µm
- Automated data collection
- Automated data processing



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Summary of the First Trip By The Numbers

- 16 hrs Rapid Access on FMX 17-2
- Shipped 1 dewar → 7 pucks → 112 samples
- Test User Groups *(thank you for providing crystals! We need more!)*
  - VU → 2 groups → 27 samples
  - UW-Madison → 1 group → 28 samples
  - WSU → 1 group → 7 samples
  - GVSU → 1 group → 50 samples
- Screened 105 samples
  - 3 non-SPINE bases
  - 4 Did Not Mount (replicates)
- 97 Datasets collected
- 1.3TB of data

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How long it takes to screen samples and collect simple datasets; grid or helical scans, MAD, etc., take longer:

- Info from Vivian Stojanoff/NLSL-II: "Proficient data collectors ~1 puck/hour"
- Joe's NLSL-II in person experience –
- Pam's 1st NLSL-II Remote experience – 50 crystals in ~6 hours
- NSLS-II – datasets take ~20s, screening takes a few minutes/crystal
- Info from SOLEIL Beamline Scientists:
  "Industrial teams 1 puck/hour, Academic users average ½ puck/hour"

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What else can happen at the other lightsources?

- Automated data collection sometimes possible
  - Pros and Cons
  - Faster than non-automated
  - Doesn't really look for the 'best' spot on your crystal
- Automated data processing
  - Convenient for a first pass evaluation on the fly
  - You can always reprocess

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How beamtime will be managed

- Beamtime will be shared equitably, the PIs will be in charge of this as they are now
- All experiments will be conducted remotely
- Beamtime will be a small fraction of what we are used to and must be run efficiently
- Some synchrotrons want the minimum number of users contacting Staff for issues, prefer communications from Joe/Pam, and have us communicate with LS-CAT Users
- Initially data will be collected by LS-CAT Beamline Scientists
- NSLS-II suggests one 'Project Champion' per laboratory who is a proficient data collector
- Having crystals ready to go and sent to LS-CAT will be a big part of this scenario, getting your crystals to LS-CAT earlier than you have a scheduled spot is ideal
- If you have no crystals for your scheduled time, you will be rescheduled

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Beamtime already scheduled thus far --

Upcoming beamtime (16 hrs/experiment) for which we are looking for volunteers to offer samples while APS is still operating are:

3/2 ESRF – ID30A-3	(Joe On-Site)
3/13 NSLS-II – FMX	(need samples at LS-CAT by 3/7)
3/25 NSLS-II – AMX	(need samples at LS-CAT by 3/21)
3/31 ESRF – ID23-1	(need samples at LS-CAT by 3/28)

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Beamtime already scheduled thus far --

This time scheduled during the Dark Period. More will be scheduled at NSLS-II for their next cycle which begins in May (20 shifts requested):

4/20 ESRF	ID23-1	6/24 ESRF	ID23-2
4/29 ESRF	ID30B	6/29 ESRF	BM07
5/5 ESRF	BM07	7/7 ESRF 7/8 ESRF	BM07 ID30-B
5/11 ESRF	ID30B	7/13 ESRF	ID23-2
5/22 ESRF	ID30B	7/16 ESRF	ID30A-3
6/9 ESRF	ID23-1	7/20 ESRF	ID23-1
6/17 ESRF	ID30A-3	7/23 ESRF	ID23-2

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